

Root proliferation in decaying roots and old root channels: a nutrient conservation mechanism in oligotrophic mangrove forests?

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Summary

1 In oligotrophic habitats, proliferation of roots in nutrient-rich microsites may contribute to overall nutrient conservation by plants. Peat-based soils on mangrove islands in Belize are characterized by the presence of decaying roots and numerous old root channels (0.1–3.5 cm diameter) that become filled with living and highly branched roots of *Rhizophora mangle* and *Avicennia germinans*. The objectives of this study were to quantify the proliferation of roots in these microsites and to determine what causes this response.

2 Channels formed by the refractory remains of mangrove roots accounted for only 1–2% of total soil volume, but the proportion of roots found within channels varied from 9 to 24% of total live mass. Successive generations of roots growing inside increasingly smaller root channels were also found.

3 When artificial channels constructed of PVC pipe were buried in the peat for 2 years, those filled with nutrient-rich organic matter had six times more roots than empty or sand-filled channels, indicating a response to greater nutrient availability rather than to greater space or less impedance to root growth.

4 Root proliferation inside decaying roots may improve recovery of nutrients released from decomposing tissues before they can be leached or immobilized in this intertidal environment. Greatest root proliferation in channels occurred in interior forest zones characterized by greater soil waterlogging, which suggests that this may be a strategy for nutrient capture that minimizes oxygen losses from the whole root system.

5 Improved efficiency of nutrient acquisition at the individual plant level has implications for nutrient economy at the ecosystem level and may explain, in part, how mangroves persist and grow in nutrient-poor environments.

Key-words: flooding, mangrove wetland, morphological plasticity, nutrient efficiency, peat, phosphorus, root proliferation, swamp, tropical forest

Journal of Ecology (2001) **89**, 876–887

Introduction

Phenotypic plasticity allows plants to adjust during ontogeny to spatially and temporally variable resources. Root plasticity, in particular, is thought to be a central feature of plant adaptation to the environment, allowing dynamic response to variation in soil nutrients or moisture (Grime 1994). Plants are also thought to partition resources optimally to achieve capture of limiting resources at minimal cost (Gleeson & Tilman 1992; Grime 1994). Although some species

are less responsive than others to spatial and temporal variation in nutrients (Fitter 1994), localized proliferation of roots in nutrient-rich patches may be an example of an 'optimal foraging' response. The most dramatic examples come from crop species, in which increased root growth and branching occur in fertilized bands (e.g., Drew 1975). The majority of studies that experimentally examined root proliferation in nutrient-rich zones have been conducted under controlled environmental conditions, either in solution culture or in sand amended with inorganic solution (see Robinson 1994 for a review). Only a few studies have examined root responses to nutrient-rich organic matter added in patches, a situation more closely resembling the natural

environment (van Vuuren *et al.* 1996; Wijesinghe & Hutchings 1997; Hodge *et al.* 1998; Robinson *et al.* 1999). Examples from natural plant communities are relatively rare compared to those from agricultural systems (Robinson 1994). Field studies, however, have shown variation in root proliferation in response to seasonality and phenological stage (Eissenstat & Caldwell 1988), neighbouring plant roots (Caldwell *et al.* 1991), duration and timing of nutrient patches (Pregitzer *et al.* 1993), litter accumulation in topographic depressions (Roy & Singh 1994) and shading and plant nitrogen status (Bilbrough & Caldwell 1995).

Although field investigations may not fully reveal underlying mechanisms for root proliferation, they allow insight into natural patterns of root growth and distribution where nutrients are spatially and temporally variable. Examination of natural plant communities, particularly those that dominate nutrient-poor habitats, can often provide clues about the ecological advantages of phenotypic plasticity via processes such as root proliferation. Here, I study the proliferation of fine, highly branched mangrove roots that occur within the decomposing roots and old root channels of these trees on intertidal islands in the Belizean Barrier Reef Complex. These mangrove-dominated ecosystems are remote from the influence of terrigenous sediments and anthropogenic inputs of nutrients, and their close proximity to coral reef and seagrass communities illustrates the oligotrophic nature of this environment (Rützler & Feller 1996).

These oceanic island systems, which are common throughout the Caribbean, are fundamentally different from mangrove forests occurring along muddy coasts, where high inputs of mineral sediments provide a substantially greater nutrient supply (Rützler & Feller 1999). In such nutrient-poor habitats, mangroves must rely on efficient nutrient acquisition, resorption and recycling mechanisms to persist and grow (Feller 1995; Feller *et al.* 1999), as well as cope with high salinity (> sea strength), periodic flooding by tides, unstable sediments, rising sea level and hurricanes (Macintyre *et al.* 1995; McKee 1995; Woodroffe 1995). Mangroves are well known for a suite of adaptations allowing their establishment, growth and reproduction in such an inimical habitat (Tomlinson 1986). In particular, the structure and function of mangrove roots have received much attention (e.g. Chapman 1940; Gill & Tomlinson 1975; McKee *et al.* 1988), but their proliferation has only been reported in encrusting sponges (Ellison *et al.* 1996).

The objective of this study was to quantify the occurrence of mangrove roots within decomposing roots and old root channels and to determine what factors may stimulate proliferation in these microsites. Three alternative hypotheses were tested: (1) the root channels decrease impedance to, or increase space for, root growth, (2) a non-resource stress condition (e.g. soil redox potential) present in the bulk soil is ameliorated within the root channels or (3) the channels provide access to decomposing organic matter and nutrients.

This work was carried out at Twin Cays, a 91.5 ha island range, which is the Smithsonian Institution's primary field site for mangrove research in Belize (Rützler & Feller 1996). The results contribute to a better understanding of how plant roots may 'forage' for limiting nutrients in a natural setting and how mangroves, in particular, manage to survive in a flooded, low-nutrient environment.

Methods

STUDY SITE

Twin Cays is located 2.3 km west of the barrier reef crest and 12 km from the mainland in central Belize (16°50'N, 88°06'W). According to corings and radio-carbon dating, Twin Cays was established on a Pleistocene limestone base *c.* 7000 years ago and has built vertically through peat accumulation (Macintyre *et al.* 1996). Freshwater, terrigenous sediment and nutrients from coastal runoff do not reach Twin Cays because of its distance from the mainland. Marine sediment, primarily in the form of calcium carbonate sand or marl, accumulates in thin lenses around the periphery of the islands, but the predominant substrate is peat derived from mangrove tissues. The tides in this region are microtidal (mean range = 15 cm; spring range = 18.4 cm; maximum range = 50 cm) and of a mixed semidiurnal type (Kjerfve *et al.* 1982). Porewater salinities vary from 30 to > 75‰, depending on rainfall, topography and evapotranspiration (McKee 1995).

Twin Cays is composed of two large islands separated by a meandering creek (0.5–2.0 m deep) with two smaller islands at the northern end of the range. The vegetative and edaphic conditions are typical of mangrove systems elsewhere in the Caribbean (McKee 1995; Woodroffe 1995). Floristically simple zones of vegetation occur along elevational gradients from the sea to the island interior. The dominant macrophyte is *Rhizophora mangle* L. (red mangrove), which forms dense stands along the shoreline with canopies averaging 3–7 m in height. Immediately landward of the red mangrove fringe and at slightly higher elevations there are mixed stands dominated by *Avicennia germinans* (L.) Stearn. (black mangrove), with *R. mangle* often occurring as a subdominant. *Laguncularia racemosa* (L.) Gaertn.f. (white mangrove) occurs infrequently in the mixed zone at this site. Extensive stands of scrub *R. mangle* (1.5–3 m in height) occur farther inland, but at lower elevations, whereas dwarf *R. mangle* trees (< 1.5 m) occupy shallow ponds and flats in the island centre.

HYDROEDAPHIC CONDITIONS

To characterize the rooting environment in each of the three forest zones, tidal fluctuation and several soil variables affected by tides were measured. Wells (7 cm in diameter and 30 cm deep) were cored at *c.* 5 m intervals along two transects from shore to scrub. Water level

relative to the soil surface was measured during neap tide to determine relative water depths at high and low tides as well as degree of soil flushing. Soil redox potential (E_h) and porewater salinity, pH and concentrations of sulphide, PO_4 -P and NH_4 -N were determined as described previously (McKee *et al.* 1988; McKee 1995).

DISTRIBUTION OF ROOTS AND ROOT CHANNELS

Three replicate cores (7 cm diameter $\times$ 2.5 m deep) were extracted in 50 cm sections with a Russian peat corer from each of the three forest zones (fringe, mixed and scrub). The cores were divided into 10 cm sections, which were dissected to determine the depth distribution of root channels and live roots. Shallow peat cores (16 cm diameter $\times$ 20 cm depth) were extracted from each zone ($n = 10$) and sliced lengthwise to determine root channel characteristics and root occurrence in channels. The diameters of dead roots and root channels visible on a single, vertical face of the divided core were measured. Live roots growing inside root channels were separated from those growing in the bulk soil. Roots were considered to be alive (or recently so) if they were light in colour, turgid and structurally intact and could also float (indicating that internal air spaces were not filled with water). Roots growing in the fringe and scrub zones were of a single species (*R. mangle*); roots in the mixed zone (*R. mangle* and *A. germinans*) were separated by species based on macro- and microscopic characteristics (McKee, unpublished data). It is important to note that mature, below-ground roots of *R. mangle* and *A. germinans* exhibit little secondary growth and retain the epidermis (lignified) and lacunose cortex; the stele typically has a porous medulla (Gill & Tomlinson 1975; K. McKee, personal observation).

Redox potentials were also measured inside root channels and compared to simultaneous measurements in the bulk soil adjacent to the channel to determine whether intensity of soil reduction was ameliorated inside channels. Nine soil cores (7 cm in diameter) were extracted from the scrub zone and extruded onto a plastic sheet. Root channels (1–2 cm in diameter) that completely traversed the width of the core, that were filled with organic matter, and that did not contain in-grown roots, were identified. The tip of a platinum electrode (c. 3 mm in diameter) was carefully inserted c. 2 cm into one randomly selected root channel per core, and a duplicate electrode was inserted to the same depth in the bulk soil adjacent to the channel. Channels containing live roots were not used because oxygen leakage from in-grown roots may alter E_h (McKee *et al.* 1988).

EXPERIMENT WITH ARTIFICIAL ROOT CHANNELS

Artificial root channels were created with sections of PVC pipe (2 cm inner diameter $\times$ 18 cm length) that

simulated the larger channels formed by *R. mangle*. The pipe sections were open at the ends and were drilled along their lengths with numerous holes (0.6 cm diameter). The pipe sections were filled with two types of root-free sediment common to mangrove island forests in Belize. The first was fine sand, which is formed from the calcareous alga *Halimeda* sp. and accumulates along the leeward sides of islands. However, in the interior of Twin Cays, thick mats of organic matter produced by algae and bacteria accrete on the surface of unvegetated flats. Such organic deposits in which the original plant parts cannot be identified, as opposed to peat comprised of slightly decayed or non-decayed plant parts that are readily identifiable, are termed muck (Brady 1984).

Two sets of PVC pipes were filled with either sand or muck, both of which had been hand-sorted to remove leaves, roots and wood, and a third set remained empty to provide only space for root growth. Replicate sets of pipe were placed in the fringe and interior scrub stands dominated by *R. mangle*. Fifteen holes (2.5 cm diameter) were cored in the mangrove peat at each site along a line parallel to the shoreline, and the pipes were positioned in a randomized design. The pipes were inserted vertically with the tops positioned 3 cm below the soil surface, and the empty space above was plugged with a section of the removed core. The pipes were retrieved after 2 years and the contents were washed over a 0.1 mm sieve. Roots were separated into fine (≤ 2 mm) and coarse (> 2 mm) size fractions, dried at 70 °C and weighed.

Various physical and chemical properties of sand and muck substrates were measured on subsamples of these materials for comparison with the bulk peat soil. Undisturbed, unvegetated deposits of sand and muck were identified at the shoreline and on unvegetated flats, respectively, and a randomly selected subset was sampled. A set of random peat samples was similarly collected from vegetated areas traversing fringe, mixed and scrub mangrove zones. Soil redox potentials were determined *in situ* and porewater was collected for measurement of pH and sulphide (McKee *et al.* 1988). Subsamples of the sand and muck used to fill the artificial channels were collected with a piston corer for determination of bulk density, percent ash and organic matter contents (Karam 1993; Parent & Caron 1993), total nitrogen (TN) (Perkin-Elmer CHNS analyser, #2400 Series 2) and total phosphorus (TP) (acid digestion; Murphy & Riley 1962; APHA 1992).

STATISTICAL ANALYSIS

The data were analysed by one-way or two-way ANOVAS, where zone (fringe, mixed, scrub) or zone and treatment (sand, muck, empty), respectively, were grouping factors. Any data that did not meet the assumptions of variance homogeneity or normality were transformed and retested prior to analysis. In general, variance was reduced for continuous data by

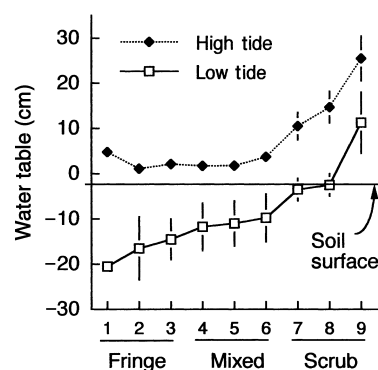


Fig. 1 Water level was measured relative to the soil surface at each station at low and high water (neap tide) across vegetation zones. Station 1 was located at the shoreline, and succeeding stations occurred at *c.* 5 m intervals along transects perpendicular to the shoreline. Values are the mean $\pm$ SE ($n = 2$); note that some SE bars are smaller than symbols.

log transformation and for non-continuous data (counts) by square-root transformation. Proportions were arcsine transformed prior to analysis. A chi-square goodness-of-fit test was also used to assess occurrence of roots inside channels as: $\chi^2 = \sum (O - E)^2/E$, where O = the observed mass of roots inside channels and E = the expected mass of roots inside channels based on the proportional volume of soil occupied by channels. Statistical tests were performed with JMP® for the Macintosh® (SAS JMP 1998).

Results

HYDROEDAPHIC CONDITIONS

Water levels relative to the soil surface varied over a tidal cycle and with distance from the shoreline (Fig. 1). Depth of the water table at low tide in the scrub zone was significantly different ($F = 11.39$, $P < 0.001$) from that in the fringe and mixed zones, which were not significantly different. Along the shoreline and in the mixed zone, the soil was drained at low tide to depths of 17.2 ± 2.5 and 10.8 ± 3.6 cm, respectively, whereas at high tide water depths above the soil surface were only 2.7 ± 0.7 and 2.4 ± 0.8 cm, respectively. In contrast, the scrub sites remained flooded at or above the soil surface over a tidal cycle, and water

depths at high tide were significantly higher than in fringe and mixed zones ($F = 16.88$, $P = 0.001$). Thus, even though water level fluctuated similarly over a tidal cycle in fringe, mixed and scrub zones (ranges of 19.8, 13.2 and 15.2 cm, respectively) ($P > 0.05$), the degree of soil flushing was greater in the fringe and mixed zones than in the scrub stand. Observations at other times during a lunar cycle indicate that the soil surface in scrub areas is occasionally exposed during a spring tide, but otherwise remains completely submerged.

Less soil flushing and extended hydroperiod in scrub sites generated significantly lower E_h and accumulations of sulphide compared to the other zones (Table 1). Porewater NH_4 concentrations also varied along this flooding gradient and were significantly higher in the scrub site, whereas PO_4 concentrations remained low across sites (Table 1). Although salinity and pH were also significantly higher in the scrub zone, the differences were small (Table 1). No difference in E_h was found between the root channels (-169 ± 28 mV) and the bulk soil (-231 ± 73 mV) (paired t -test, $P > 0.05$).

DESCRIPTION AND DISTRIBUTION OF ROOTS AND ROOT CHANNELS

Each of the 2.5 m deep cores exhibited a similar profile: an upper layer (5–10 cm thick) comprised of mostly living, fine roots on top of deeper layers of peat that were moderately decomposed (H5 on the Von Post Scale). Within the peat layer were numerous channels created by the persistent remains of dead mangrove roots (Fig. 2a). Most of the root channels were distinctly delineated by the refractory epidermis of below-ground roots and prop roots of *R. mangle* and cable roots and pneumatophores of *A. germinans*, which formed variously sized pores in the bulk peat.

Root channel diameter was variable, ranging from 0.1 to 3.5 cm, and reflected the size variation of each species' roots (those of *R. mangle* being somewhat larger, on average). Many channels were empty, but others contained organic matter derived from internal root structures in various stages of decomposition. Recently dead roots retained sufficient internal structure to allow recognition of the stele, aerenchymatous cortex and lignified epidermis, but the cortex of decomposing

Table 1 Summary of edaphic conditions in fringe (*Rhizophora mangle*), mixed (*R. mangle* and *Avicennia germinans*), and scrub (*R. mangle*) zones. Values are the mean $\pm$ SE ($n = 6$). Data were analysed as a one-way ANOVA with zone as the grouping factor. F -ratios are given for each variable, and significant differences are indicated by ** $P = 0.01$, *** $P = 0.001$, or n.s. (not significant)

Variable	Fringe	Mixed	Scrub	F -ratio	P
E_h at 1 cm (mV)	100 ± 35	179 ± 64	-162 ± 126	17.6	***
E_h at 15 cm (mV)	-61 ± 28	23 ± 27	-170 ± 10	17.5	***
Salinity (‰)	34 ± 1	36 ± 1	39 ± 1	7.42	**
pH	6.3 ± 0.1	6.4 ± 0.1	7.4 ± 0.1	172	***
Sulphide (mM)	0.09 ± 0.05	$< 0.01 \pm 0.00$	1.72 ± 0.32	29.5	***
$\text{NH}_4\text{-N}$ (μM)	2.9 ± 1.2	1.3 ± 0.2	15.1 ± 7.3	20.0	***
$\text{PO}_4\text{-P}$ (μM)	6.7 ± 1.6	6.0 ± 2.3	5.0 ± 1.3	0.401	n.s.

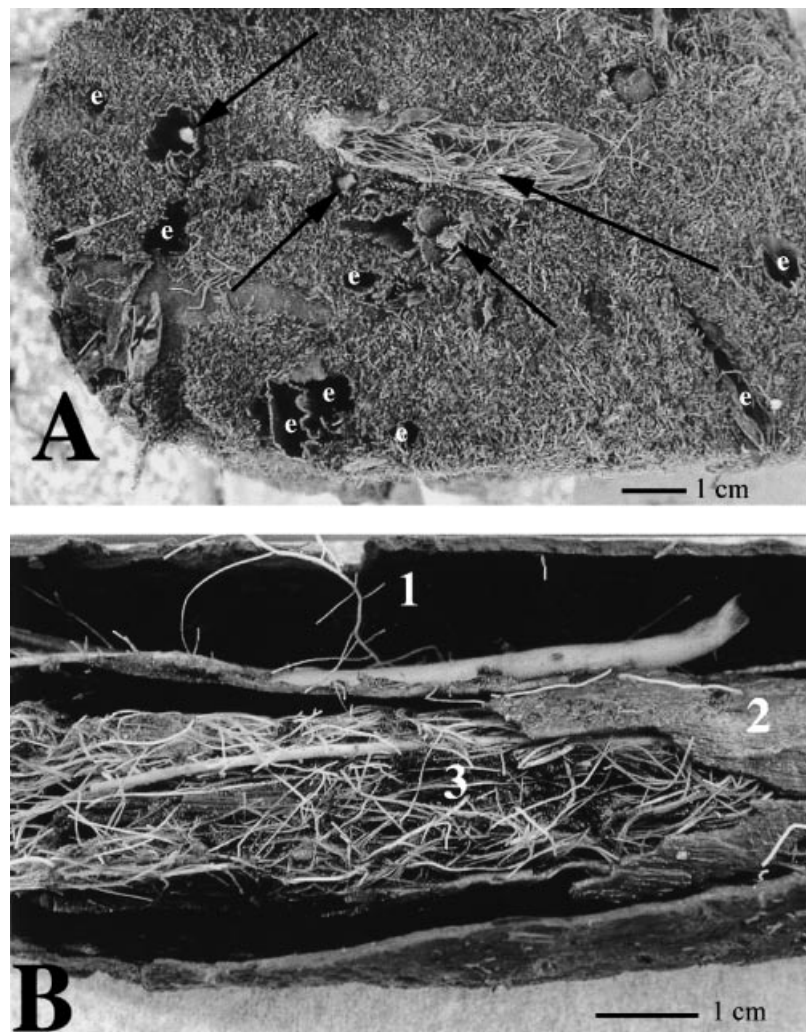


Fig. 2 Selected views of mangrove root channels with in-grown, living roots. (a) Empty root channels (e) are visible as holes in the peat matrix and are often lined with the refractory epidermis of the original root. Occupied channels with ingrown roots are indicated by arrows. (b) A double root channel, dissected from the peat and split open lengthwise, shows three generations of roots that have successively grown inside each other. The outer channel is formed by the remains of a *Rhizophora mangle* prop root (1); the inner channel, formed by a large underground root of the same species (2), is filled with fine, live roots (3) that have proliferated in the decaying organic matter.

roots was usually fragmented. Live and recently dead roots and old root channels were apparent throughout the mangrove peat profile down to a depth of *c.* 1 m at Twin Cays. However, the majority of channels and live roots was in the upper 50 cm. Dead roots and channels became increasingly fragmented with depth and were often collapsed below a 0.5 m depth.

Root channels were formed by both of the dominant mangrove species at Twin Cays and were found in monospecific as well as mixed stands. Most of the larger channels were readily identifiable to species by comparison with living and recently dead roots collected in monospecific stands. In the case of *R. mangle*, the channels formed by the lignified epidermis and outer cortex were often dark red and > 1 cm in diameter, making identification relatively easy. Roots of *A. germinans* were smaller on average, and only the cable roots and pneumatophores (pencil-like aeration structures *c.* 1 cm diameter) appeared to form distinct channels. *Avicennia*

germinans channels were also formed primarily by the epidermis, which was typically grey to black in colour. Other macro- and microscopic features differentiating the species were found (data not presented), but were only occasionally needed to distinguish channel origin. For example, the aerial roots of *R. mangle* formed channels if they became buried by sediment, and these were distinguished from below-ground roots by their larger diameter (> 2 cm) and a thickened corky layer with lenticels (openings for gas exchange) (Fig. 2b).

Another major difference between channels formed by the two species was in their orientation, which reflected the growth pattern of the two species' root systems. The cable roots of *A. germinans* typically formed a horizontal, underground network radiating outward from the base of the tree at *c.* 20–30 cm depth. Pneumatophores occurred at intervals along the cable root and grew upwards out of the soil. The root system of *R. mangle* also spread horizontally from the main trunk,

Table 2 Summary of root channel characteristics measured in fringe (shoreline) and scrub (interior) zones dominated by *Rhizophora mangle*. Measurements were made on all channels visible on one interior face of each core, halved lengthwise (16 cm diameter $\times$ 20 cm depth). Values are the mean $\pm$ 1 SE ($n = 10$ cores). Data were analysed as a one-way ANOVA with zone as the grouping factor. F -ratios are given for each variable, and significant differences are indicated by * $P = 0.05$, n.s. (not significant)

Variable	Fringe	Scrub	F -ratio	P
Density (number m^{-2})	156 $\pm$ 23	219 $\pm$ 19	4.39	*
Diameter (cm)	0.73 $\pm$ 0.10	0.80 $\pm$ 0.08	0.366	n.s.
Cross-sectional area ($cm^2 m^{-2}$)	122 $\pm$ 37	188 $\pm$ 26	2.08	n.s.
Percent (%) of total soil volume	1.2 $\pm$ 0.4	1.9 $\pm$ 0.3	2.07	n.s.

Table 3 Number and diameter of living, primary roots growing inside channels formed by dead *Rhizophora mangle* or *Avicennia germinans* roots in the fringe, mixed and scrub zones. Measurements were made on peat monoliths (16 cm $\times$ 16 cm $\times$ 20 cm). Data were analysed as a two-way ANOVA with zone and channel-forming species (nested within zone) as the grouping factors. F -ratios are given for each variable, and significant differences are indicated by * $P = 0.05$ or n.s. (not significant)

Zone	Root channel		In-grown roots		Root diameter (mm)
	Species	Total	Total	Mean $\pm$ SE	
Fringe	<i>Rhizophora</i>	11	64	8.0 $\pm$ 1.7	3.4 $\pm$ 0.3
Mixed	<i>Rhizophora</i>	5	23	4.6 $\pm$ 1.4	2.4 $\pm$ 0.4
	<i>Avicennia</i>	5	11	2.2 $\pm$ 0.6	3.0 $\pm$ 0.7
Scrub	<i>Rhizophora</i>	13	88	6.8 $\pm$ 1.0	3.2 $\pm$ 0.3
ANOVA source					
Zone				4.11*	0.792 n.s.
Species (within Zone)				1.10 n.s.	0.521 n.s.

but did so above ground through the production of a series of arching, stilt roots. Below-ground roots of *R. mangle* grew at various angles relative to the soil surface, e.g. horizontally just under the soil surface or vertically to depths of 1–2 m. Channels formed by these different root structures thus created a variety of pores that varied in size, orientation and depth distribution within the peat matrix.

Live roots of both *R. mangle* and *A. germinans* grew inside decaying roots and root channels of both species, generally following the direction of the old root until it became fragmented or collapsed (Fig. 2a,b). The living roots occupied the cortical space, often penetrating the spongy tissue in decaying roots. Many root channels in all sample sites were unoccupied, whereas others contained one or more live roots that grew lengthwise through the channel. Occupied channels were often packed solid with primary roots and their lateral roots. Occasionally, a single large root occurred inside a root channel and filled the entire space, its shape often distorted by the dimensions of the root channel. In some cases, a large root 'outgrew' its channel, splitting it open and leaving only strips of the epidermis that originally formed the channel. Many of the larger root channels (> 2 cm diameter) were occupied by successive sets of roots, each growing inside the previous one (Fig. 2b).

The occurrence of root channels in mangrove peat appeared to be widespread. Cores taken at other locations in Belize's reef complex (e.g. Turneffe Atoll, Tobacco Range and the Pelican Cays) also contained numerous root channels filled with live mangrove roots (data not presented).

ABUNDANCE OF ROOT CHANNELS AND IN-GROWN ROOTS

All shallow root cores (20 cm deep) collected for quantification of root channels and in-grown root mass contained at least one channel (empty or occupied). Root channels occurred at an average density of $188 \pm 16 m^{-2}$ (range = 63–344 m^{-2}) (Table 2). Channel density was significantly higher in the scrub compared to the fringe zone (Table 2). Neither channel diameter (average 0.77 ± 0.06 cm) nor total cross-sectional area of root channels visible on the sampled core face (average $155 \pm 23 cm^2 m^{-2}$), therefore the proportion of soil volume occupied by root channels (estimated to be $c. 1.6 \pm 0.2\%$, range = 0.5–10.5%), differed between fringe and scrub sites (Table 2).

The average number of primary roots (three) inside occupied channels in the mixed zone was significantly less ($F = 8.06$; $P = 0.01$) than the seven or eight in the fringe and scrub zones, which did not differ (Table 3). The number of roots in channels formed by *R. mangle* was slightly higher than that in *A. germinans* channels (five vs. two), but the difference was not significant. The average diameter of primary roots inside channels was 3.3 ± 0.2 mm (range = 0.6–16 mm) and did not differ among zones (Table 3). The in-grown primary roots typically had long, fine lateral roots (0.1–0.3 mm in diameter).

The proportion of live roots occurring in old root channels varied from 9 to 24% of the total root mass in a core, depending on proximity to the shoreline (Fig. 3). The live root biomass inside the channels

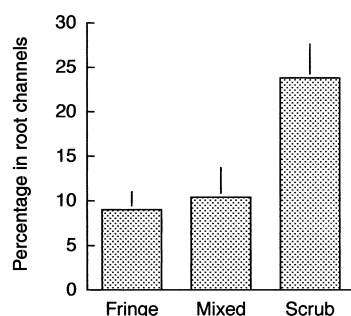


Fig. 3 Percent (%) of total live root mass occurring inside natural root channels in fringe (*Rhizophora mangle*), mixed (*R. mangle* and *Avicennia germinans*) and scrub (*R. mangle*) zones. Values are the mean ± 1 SE ($n = 11-13$).

was significantly higher in all zones separately and together than would be expected by chance, based on the proportion of soil volume occupied by root channels being 1.6% (Table 4). The proportion of roots inside channels was significantly higher in the *Rhizophora* scrub forest than in the *Rhizophora* fringe or in the mixed forest ($F = 6.97$, $P = 0.01$). In the mixed forest, *R. mangle* and *A. germinans* showed no significant difference in root occupation of old channels ($F = 0.007$, $P > 0.05$).

ARTIFICIAL ROOT CHANNELS

Physico-chemical characteristics of the sand and muck substrates used in the artificial root channels are summarized and compared to peat in Table 5. The sand provided an inorganic, nutrient-poor rooting matrix, whereas the muck provided an organic, nutrient-rich substrate. The sand was further characterized by a higher bulk density and lower total pore space compared to muck and peat. The muck exhibited lower E_h and higher sulphide concentrations compared to peat and sand (Table 5).

The artificial channels filled with organic muck had six times more root biomass than the empty or sand-filled channels ($F = 14.6$, $P = 0.001$), which were not significantly different (Fig. 4). After 2 years, the amount of roots in the muck-filled pipes ranged from 0.93 to 1.40 g d. wt in scrub and fringe sites, respectively, compared to 0.11–0.34 g d. wt in empty or sand-filled pipes. Averaged over treatment, root mass was greater in the fringe (0.71 g d. wt) than in the scrub (0.44 g d. wt) zone ($F = 3.69$, $P = 0.072$), and there was no significant site by treatment interaction ($F = 0.992$, $P > 0.10$). Most of the in-grown roots were 0.1–0.3 mm in diameter, although large (> 1 cm diameter) roots also occurred inside the muck-filled pipes. Only one of the

Table 4 Chi-square table showing total mass of live roots in peat cores, that observed in channels, and that expected in channels (based on percent of peat core volume occupied by channels) in fringe, mixed and scrub zones. The χ^2 statistic was calculated based on absolute root mass (g) measured in cores and channels, but the values (mean ± 1 SE) are given in g m^{-2} (to a 20 cm depth). A χ^2 statistic that exceeds the $\alpha = 0.05$ critical value requires rejection of the null hypothesis that the expected and observed root masses in channels are equal

	Fringe	Mixed	Scrub	Overall
Total in core	311 $\pm$ 62	153 $\pm$ 24	147 $\pm$ 25	199 $\pm$ 25
Expected in channels	3.7 $\pm$ 0.7	2.4 $\pm$ 0.4	2.8 $\pm$ 0.5	3.0 $\pm$ 0.3
Observed in channels	33 $\pm$ 10	11 $\pm$ 3	29 $\pm$ 4	24 $\pm$ 4
χ^2	762	212	504	1478
d.f.	9	10	11	34
Critical value ($\alpha = 0.05$)	3.33	3.94	4.57	c. 22

Table 5 Physico-chemical properties of three types of soil substrate found at Twin Cays: calcareous sand, muck and fibrous peat. Soil redox (E_h) measurements were made *in situ*, and samples of porewater were collected for determination of pH and sulphide. Samples of each substrate were collected from an adjacent undisturbed area with a piston corer for determination of bulk density, percent ash and organic matter, and total nitrogen (N) and total phosphorus (P). Values are the mean ± 1 SE ($n = 6$). F -ratios are given for each variable, and significant differences are indicated by ** $P = 0.01$, *** $P = 0.001$ or **** $P = 0.0001$

	Sand	Muck	Peat	F
Bulk density (g cm^{-3})	1.40 $\pm$ 0.06	0.10 $\pm$ 0.01	0.14 $\pm$ 0.001	500***
Ash (%)	96 $\pm$ 1	36 $\pm$ 1	28 $\pm$ 1	2085***
Organic matter (%)	4 $\pm$ 1	63 $\pm$ 1	73 $\pm$ 1	2075***
Total pore space (%)	45 $\pm$ 3	94 $\pm$ 1	92 $\pm$ 1	314***
Total N (mg g^{-1})	1.8 $\pm$ 0.1	24.0 $\pm$ 0.3	8.0 $\pm$ 0.5	1086***
Total P (mg g^{-1})	141 $\pm$ 22	406 $\pm$ 19	161 $\pm$ 6	79.9***
E_h at 1 cm (mV)	114 $\pm$ 31	-236 $\pm$ 25	83 $\pm$ 1	69.3***
E_h at 15 cm (mV)	-17 $\pm$ 20	-173 $\pm$ 25	41 $\pm$ 21	24.9***
pH	6.64 $\pm$ 0.11	6.92 $\pm$ 0.11	6.12 $\pm$ 0.18	8.37**
Sulphide (mM)	0.06 $\pm$ 0.03	2.30 $\pm$ 0.13	0.33 $\pm$ 0.21	74.8***
Salinity (‰)	35 $\pm$ 1	36 $\pm$ 1	45 $\pm$ 4	6.45**

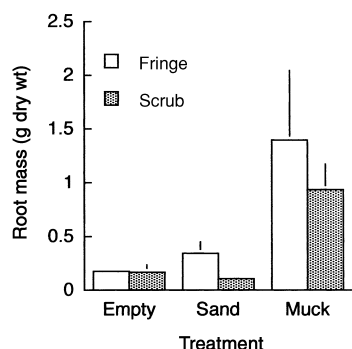


Fig. 4 Root biomass in artificial root channels constructed of PVC pipe and buried in fringe and scrub zones dominated by *Rhizophora mangle*. The pipes, which were filled with calcareous sand, organic muck or left empty, were retrieved after 2 years. Values are the mean ± 1 SE ($n = 3-5$).

empty or sand-filled channels contained a large root (1.5 cm diameter $\times$ 2 cm length), which had grown horizontally through two opposing holes and had expanded inside the pipe. In contrast, half of the muck-filled channels had large roots (> 1 cm diameter) growing lengthwise through the entire pipe. All of the large roots found inside the artificial channels were alive. Fine roots (< 2 mm in diameter), which were not separated by condition, accounted for $54 \pm 8\%$ of the total in-grown root mass in muck-filled channels.

Discussion

POTENTIAL CAUSES OF ROOT PROLIFERATION IN OLD ROOT CHANNELS

The mangrove trees at Twin Cays and throughout the Belize Barrier Reef Complex clearly exhibited root proliferation inside decaying roots and old root channels. Root channels may provide a path of least resistance through an impermeable matrix, as observed in some compacted soils (Heilman 1981; Dell *et al.* 1983; Sollins & Radulovich 1988). However, the peat soil in which the mangrove roots were growing was relatively soft, and corings did not reveal any compacted or impenetrable layers. In fact, the coring device was pushed by hand to a 2.5 m depth, well beyond the rooting zone. Furthermore, deeper corings indicate that the limestone bedrock lies 8–10 m below the soil surface (Macintyre *et al.* 1986).

Another possibility is that the root channels simply provide more space for growth, i.e. larger pores than the interstitial spaces in the peat, and therefore less growth restriction. However, the artificial root channels that were empty did not have more roots than the ones that were filled with sand, confirming that additional growing space did not generate greater root proliferation by mangroves (Fig. 4). Also, total pore space in surface mangrove peat at Twin Cays was high (Table 5).

The root channels could have also ameliorated some stress factor generally present in the soil at Twin Cays. Mangrove soils are often characterized by high salinity,

low E_h (indicating a strong soil oxygen demand) and accumulation of sulphide, a known phytotoxin that affects mangrove growth (McKee *et al.* 1988; McKee 1993; McKee 1995; Table 1 this study). Although salinity was not measured directly in the root channels, it could not have been much lower than that measured in the surrounding peat (Table 1), which was similar to that of seawater (c. 36‰) that floods the soil twice daily (Fig. 1). E_h measurements also indicated no difference in redox status between unoccupied root channels and the soil, but this comparison does not completely eliminate the possibility. The selected channels may have been unoccupied because they did not provide more beneficial conditions than the bulk soil. However, even if occupied channels had been measured and found to have higher redox status and/or lower sulphide, a causal relationship could not be assumed because of the reciprocal effects of root oxygen leakage on these soil factors (McKee *et al.* 1988; McKee 1993). Furthermore, the fact that root production was increased in artificial channels containing muck, which was characterized by strongly reducing conditions and high sulphide (Fig. 4, Table 5), suggests that these factors do not always inhibit root proliferation. However, they may play a role in root morphology, as discussed below.

The most reasonable explanation is that the roots proliferated inside decaying roots upon contact with decomposing organic matter and nutrients. Plant root systems generally respond to soil heterogeneity by proliferation in the most nutrient- or water-rich zones (Fitter 1994), and numerous studies have experimentally demonstrated increased branching and fine root production in nutrient-enriched patches (Robinson 1994; Roy & Singh 1994; Hodge *et al.* 1998). Mangrove root proliferation in response to nutrient-rich organic matter, but not to greater space or a reduction in physical impedance to root growth in artificial channels (Fig. 4), is consistent with this explanation.

The mechanism whereby roots proliferate in decaying organic material or areas of increased nutrient concentrations is most likely through increased branching by a root randomly encountering the richer area, rather than by a locational mechanism (i.e. a tropic response to an environmental cue such as a nutrient gradient) (Fitter 1994). The pattern of mangrove root growth in natural channels agreed with this explanation. Many natural channels contained no in-grown roots, and in occupied channels the roots grew along the longitudinal axis until diverted by a rupture or collapse of the channel wall. Also, the muck-filled artificial channels contained primary roots with many lateral roots, suggesting that proliferation occurred as a consequence of increased branching when appropriate conditions were encountered.

The roots of other tropical tree species growing in low-nutrient habitats are often physically associated with decomposing organic matter (Jordan & Escalante 1980; St. John 1983; St. John *et al.* 1983; Roy & Singh 1994). Some tropical trees exhibit extensive root mats

in the leaf litter layer on the soil surface (Jordan & Escalante 1980; St. John *et al.* 1983; Roy & Singh 1994). Mangroves may also produce roots within the leaf layer when present (K. McKee, personal observation), but some mangrove forests lack a litter layer due to physical removal by tides or consumption by detritivores (Robertson 1991). However, mangrove root proliferation into other microsites has been reported. Red mangrove trees growing along the shoreline of Belizean islands show root growth into sponges encrusting the subtidal portions of their prop roots (Ellison *et al.* 1996). Relative abundances of ^{15}N ($\delta^{15}\text{N}\text{‰}$) and ^{13}C ($\delta^{13}\text{C}\text{‰}$) suggest that these mangrove roots obtain dissolved inorganic nitrogen from sponges, and that sponges obtain carbon from mangrove roots. This facultative mutualism may enhance mangrove growth, but only trees positioned along the island periphery and edges of tidal creeks (where their roots extend into the subtidal zone) could potentially benefit from this relationship. The majority of mangrove trees on Belizean cays must derive nutrients from other sources. Because terrigenous inputs of sediment are negligible, an important external source of nutrients is seawater. However, concentrations of nitrogen and phosphorus in seawater are low (Davis 1986). The calcium carbonate sand derived from calcareous algae (e.g. *Halimeda* spp.) that accumulate along the leeward shorelines also contains little N and P (Table 5). The largest store of nutrients occurs below ground in the peat substrate, but much of this may be in refractory forms, as in other peat-forming wetlands (Richardson 1999) and therefore unavailable for plant uptake. Also, plant roots must compete with bacteria and algae for nutrients. The proliferation of roots inside decaying roots places absorbing structures in intimate contact with nutrients released during mineralization before they can be leached by tidal flushing or immobilized by chemical or biological reactions.

Do species occupying the same habitat exhibit similar nutrient foraging patterns and does root proliferation in nutrient-rich microsites result in similar growth benefits for species? Both *R. mangle* and *A. germinans* exhibited comparable root proliferation in channels, which suggested similar foraging strategies. Einsmann *et al.* (1999), however, found that root proliferation in nutrient-rich patches was highly variable among 10 co-occurring plant species (coastal plain of South Carolina) examined in terms of scale (root mass and root length densities), precision (preferential proliferation in nutrient-rich vs. less fertile patches) and sensitivity (growth benefits gained by proliferation in patches). They further found no relationship between sensitivity and precision when plants were grown under different nutrient distribution treatments in pots, indicating that proliferation of roots in fertile patches does not always result in growth benefits. However, there may be benefits of nutrient foraging in a natural setting, where several environmental stress factors interact to limit plant growth.

TRADE-OFF BETWEEN NUTRIENT ACQUISITION AND FLOOD TOLERANCE STRATEGIES

The larger proportion of live roots inside natural channels in the scrub vs. fringe or mixed zones suggests an interaction between nutrient availability and some other soil condition associated with tidal flooding. The development of more reducing soil conditions in interior scrub sites (Table 1) may limit development of fine lateral roots due to internal aeration requirements. A common strategy exhibited by wetland plants, including mangroves, is internal aeration accomplished through the development of aerenchyma, which provides a continuous, intercellular pathway for the rapid flux of oxygen through tissues growing in an anaerobic soil (Armstrong *et al.* 1991). However, roots of mangroves and other wetland species readily leak oxygen to the surrounding soil (McKee *et al.* 1988; Armstrong *et al.* 1991; Sorrell 1994) and if this loss is excessive, internal oxygen deficiencies and reduced energy status may develop (McKee & Mendelsohn 1987; McKee 1996). As root surface area can affect the rate of oxygen leakage (Sorrell 1994), long, thin roots with many fine laterals would be disadvantageous in a waterlogged soil with a high external oxygen demand, especially where the potential gain of nutrients was low. The maintenance of high internal oxygen levels to support rapid root extension would promote a 'foraging' growth pattern in a reducing soil environment until the root encountered a relatively nutrient-rich microsite. Then, the switch to an 'exploitation' growth pattern would be worthwhile, even if accompanied by greater oxygen leakage.

Where tidal flushing aerates the soil and brings more nutrients into contact with roots, root branching and fine root production may be maximized, e.g. in the fringe zone (Fig. 1, Table 1). In contrast, the relatively stagnant conditions in the scrub forest probably limit mass flow of nutrients to root surfaces and simultaneously generate a high external oxygen demand. Such conditions would require a trade-off between morphologies supporting internal aeration, i.e. short, thick roots with few laterals, vs. nutrient acquisition, i.e. long, thin roots with many fine laterals. Thus, although an overall increase in root system surface area by lateral branching might improve nutrient acquisition by mangroves, proliferation of roots only where there is a greater potential for nutrient capture would minimize oxygen losses from the whole root system. Not only should total uptake be increased by this response, but uptake per unit of carbon partitioned to the root system should be higher. In contrast, production of an abundance of roots randomly distributed through the bulk soil might achieve a similar total uptake, but the carbon and energy required below ground may be much higher. This hypothesis requires further testing, but the significantly greater proportion of live roots in natural channels formed in the scrub zone is consistent with this explanation.

PHOSPHORUS LIMITATION AND ROOT PROLIFERATION

A final consideration is what nutrient element might be responsible for root proliferation in channels? Just because a root system is in contact with the nutrients in the soil solution does not mean that it has equal access to all nutrients. Strongly adsorbed anions such as phosphate have diffusion coefficients as low as $10^{-9} \text{ cm}^2 \text{ s}^{-1}$ (Nye & Tinker 1977). By comparison, diffusion coefficients of nitrate and ammonium in dilute solution are c. 10^{-5} and $10^{-7} \text{ cm}^2 \text{ s}^{-1}$, respectively. Robinson (1996), using data from a well-known study by Drew (1975), calculated that local proliferation of lateral roots would not necessarily achieve significantly greater capture of mobile nutrients such as nitrate, but would improve capture of less mobile phosphate. Fine roots with a large surface area per unit mass would thus substantially improve acquisition of relatively immobile phosphorus from the substrate. Additionally, root depletion zones would be narrow, and narrow depletion zones mean that roots can be closely packed without competing for a common phosphate pool (Nye & Tinker 1977). Thus, if phosphate concentrations are low, as they are on mangrove islands in Belize, prolific branching of the roots may increase a tree's ability to capture this limited resource. The root proliferation in channels observed in this study, combined with the finding that red mangroves in the interior of Twin Cays are phosphorus-limited (Feller 1995; Feller *et al.* 1999), is consistent with these theoretical considerations.

ECOSYSTEM-LEVEL CONSEQUENCES OF NUTRIENT CONSERVATION

Resorption of nutrients from senescing leaves is often cited as a major conservation mechanism in several types of low-nutrient ecosystems (Aerts 1996), including mangroves (Feller 1995; Feller *et al.* 1999). Dwarf *R. mangle* trees at Twin Cays reabsorb 73% of phosphorus and 47% of nitrogen during leaf senescence (Feller *et al.* 1999), indicating that these mangroves are extremely efficient at conserving nutrients, particularly phosphorus, at an individual plant level. Proliferation of roots inside decaying roots may have a similar effect by concentrating absorbing organs where they can recapture nutrients from below-ground tissues as they decompose. In a dynamic habitat such as the intertidal zone, the ability to alter root growth to take advantage of spatial heterogeneity or temporal pulses in nutrient availability would be particularly important. Work in other types of forests has shown that soil nutrient concentrations vary spatially and temporally and that localized nutrient-rich areas may be an important source of nutrients for plants growing in heterogeneous environments (Farley & Fitter 1999).

Although both *R. mangle* and *A. germinans* exhibited root proliferation in channels, the amount of fine root production (and consequently absorbing surface

area per unit biomass) appears to be much greater for *R. mangle* (K. McKee unpublished data). The combination of root proliferation in decaying roots (this study), above-ground nutrient conservation strategies (Feller *et al.* 1999) and facultative mutualism with root-fouling sponges (Ellison *et al.* 1996), renders *R. mangle* particularly well-suited for growth in oligotrophic habitats. If root proliferation in enriched microsites improves nutrient capture, species such as *R. mangle* may be more competitive than other mangroves in low-nutrient systems, as demonstrated for some grassland species (Robinson *et al.* 1999).

Improved efficiency of nutrient acquisition as a consequence of selective exploitation of microsites at the individual plant level has implications for nutrient economy at the ecosystem level. If root growth and branching inside decaying roots improves nutrient acquisition, the need for allocation of biomass below ground may be reduced. The conserved carbon and nutrients could then be directed toward growth of shoots and aerial roots. Consequently, overall productivity of the ecosystem and its resilience to disturbance may be improved. Root proliferation in decaying roots (this study) and other nutrient-enriched microsites (e.g. epibionts; Ellison *et al.* 1996) may partially compensate for external nutrient limitations to mangrove growth as well as conserve nutrients within the mangrove ecosystem. Mangrove communities with tight nutrient cycles would tend to export smaller amounts of nutrients than those that are less conservative, and these differences would be expected to have an impact on secondary productivity in adjacent waters (Twilley 1988, 1995).

Acknowledgements

I would like to thank Klaus Rützler and the Caribbean Coral Reef Ecosystems (CCRE) program for funding and use of the Carrie Bow Cay field station in Belize. This work was partially funded by a grant from the National Science Foundation-Biocomplexity Program (DEB-9981483). Patti Faulkner assisted with soil analyses and Irv Mendelsohn, Tammy Charron, Darren Johnson, Ron Boustany and an anonymous reviewer provided comments on the manuscript. This is CCRE contribution number 572.

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Received 31 August 2000

revision accepted 1 April 2001